

# Saad Abdur Razzaq

**Passport:** FU1072703 | **Date of birth:** 05/10/2002 | **Place of birth:** Riyadh, Saudi Arabia | **Nationality:** Pakistani | **Gender:** Male | **Phone:** (+92) 3036668942 (Mobile) | **Email address:** [saadabdurrazzaq00@gmail.com](mailto:saadabdurrazzaq00@gmail.com) | **LinkedIn:** [saadarazzaq](#) | **Address:** 37620, Faisalabad, Pakistan (Home)

## • ABOUT MYSELF

Computer Science graduate and Machine Learning Engineer with a strong foundation in statistics, data architecture, and advanced algorithms. Experienced in building end-to-end ML solutions, optimizing localized deep learning models, and writing clean, scalable code. Passionate about turning complex mathematical concepts into practical, high-performance software.

## • EDUCATION & TRAINING

31/08/2020 - 31/05/2024 - FAISALABAD, PAKISTAN

### **BSC. COMPUTER SCIENCE-** NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Core Mathematical & Data Science Modules Studied:

- **Advanced Mathematics:** Linear Algebra , Calculus & Analytical Geometry , Differential Equations , Numerical Computing , Graph Theory.
- **Statistical & Computational Analysis:** Probability and Statistics , Data Mining , Design and Analysis of Algorithms.
- **Applied Intelligence:** Applied Machine Learning , Artificial Intelligence , Fundamentals of Computer Vision.

## • ACADEMIC AND WORK EXPERIENCE

24/07/2024 - CURRENT

### **Freelance Machine Learning Engineer**

Contracted by multiple fintech clients to architect and deploy end-to-end ML solutions for risk mitigation and quantitative decision support.

- **Algorithmic Trading & Forecasting:** Contracted to design and train predictive time-series models (LSTMs, Prophet) for stock price trends and volatility estimation, delivering actionable market insights for clients.
- **Credit Risk Infrastructure:** Built and integrated custom credit risk scoring frameworks using Logistic Regression, Random Forests, and Gradient Boosting, directly enhancing client loan approval accuracy and risk assessment.
- **Data Pipeline Automation:** Automated ETL pipelines using Pandas, NumPy, and SQL to streamline financial data ingestion, cleaning, and feature extraction from disparate client data sources.
- **Performance Optimization:** Maximized predictive accuracy and generalization across client datasets through rigorous hyperparameter tuning, cross-validation, and regularization techniques.

01/2024 - 06/2024

### **Business Analytics — Teaching Assistant**

- Contributed to course delivery by designing assignments, quizzes, and lab exercises while providing guidance on data-driven business techniques.
- Supported learning in statistical analysis, exploratory data analysis, data visualization, and predictive modeling using Python and visualization tools, reinforcing both conceptual understanding and practical skills.

- Conducted tutorial sessions covering foundational mathematical data structures, algorithmic complexity (Big-O notation), and statistical modeling paradigms for undergraduate students.
- Assisted students with Python-based applications in machine learning, feature engineering, and model evaluation, strengthening both instructional quality and personal expertise.

● TECHNICAL PORTFOLIO

Projects

1. **EffiScribe**: Engineered an AI clinical platform using **OpenAI Whisper** and **FastAPI** to process unstructured multi-speaker dialogues into structured text with Large Language Models.
  - *Key Technologies: Python, FastAPI, React Native, Next.js, Whisper, NLP, LLMs.*
2. **Notive AI**: Architected a cross-platform audio engine using on-device, highly quantized **TensorFlow Lite** and **Whisper** sub-models via **Kotlin Multiplatform** and **LiveKit**.
  - *Key Technologies: Kotlin Multiplatform, TensorFlow Lite, Whisper, LiveKit, Audio DSP.*
3. **AI German Language Trainer**: Implemented digital signal processing (**DSP**) via **Pydub** and **Vosk** hidden Markov models for real-time acoustic feature extraction and phoneme-level pronunciation accuracy.
  - *Key Technologies: FastAPI, React, Vosk, Pydub, Digital Signal Processing, TTS.*
4. **Algorithmic Hub**: Built a high-performance **NumPy** and **Pandas** simulator to model, map, and benchmark the space/time computational complexity of 11 graph-search heuristics.
  - *Key Technologies: Python, Pygame, NumPy, Pandas, Graph Theory, Data Structures.*
5. **AutoGrade**: Authored an Optical Mark Recognition (**OMR**) engine utilizing **OpenCV** affine transformations and **NumPy** vector-quantized arrays for high-throughput pixel density matrix evaluation.
  - *Key Technologies: Python, OpenCV, NumPy, Pandas, FastAPI, Computer Vision, OMR.*

● SKILLS

Programming Languages

Python, C, C++, C#, MATLAB, R

Data Science & ML Libraries

NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, Keras, OpenCV

Natural Language Processing (NLP)

SpaCy, Gensim, Flair, TensorFlow Hub

Databases & Cloud

PostgreSQL, MySQL, MongoDB, Docker, AWS

Web Frameworks & Visualization

FastAPI, Flask, Streamlit, Plotly, Seaborn, Matplotlib, LiveKit

● LANGUAGE SKILLS

|         | UNDERSTANDING |         | SPEAKING          |                    | WRITING |
|---------|---------------|---------|-------------------|--------------------|---------|
|         | Listening     | Reading | Spoken production | Spoken interaction |         |
| ENGLISH | C1            | C1      | C1                | C1                 | C1      |

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user